

Problem-Solution Fit

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Problem-Solution Fit Canvas

1. CUSTOMER SEGMENT(S) [CS] Loan applicants: salaried employees, self-employed individuals applying to banks/NBFCs; Bank credit officers reviewing and approving applications daily.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited tech literacy among applicants - Officers have limited time per application - No standardized scoring step before manual review	5. AVAILABLE SOLUTIONS PROS & CONS [AS] PROS: Faster decisions, data-consistent scoring CONS: Manual review is slow and inconsistent; vendor solutions are expensive
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Manual review takes days (every application, daily frequency). Inconsistent decisions across officers. Self-employed applicants assessed unfairly due to irregular income.	9. PROBLEM ROOT / CAUSE [RC] No standardized, data-driven scoring step exists before manual underwriting begins. Decisions rely entirely on officer judgment and experience.	7. BEHAVIOR + ITS INTENSITY [BE] Officers manually cross-check income, credit history, and property details for every single case — high effort, repeated daily, high intensity.
3. TRIGGERS TO ACT [TR] Applicant urgently needs funds (home/vehicle/education). Officer faces a backlog of pending loan files. Bank management demands faster turnaround and fewer NPAs.	10. YOUR SOLUTION [SL] Smart Lender: an ML-powered web application that evaluates applicant data using XGBoost (best of 4 models: Decision Tree, Random Forest, KNN, XGBoost). Applicant fills an 11-field web form → Flask backend preprocesses and scales the input → XGBoost model predicts Approved / Declined in < 1 second. Achieved ~82% test accuracy after SMOTE-based class balancing and 5-fold cross-validation.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Bank/NBFC web portal, officer internal portal, loan applicant mobile/web app interface
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Anxious about qualifying; uncertain how long it will take; worried about bias AFTER: Reassured by a fast, consistent, data-backed first-pass decision		OFFLINE In-branch visit with a loan officer, paper-based application form, manual document submission at the bank